LOW LEVEL DESIGN (LLD)

RAG-Based Healthcare Assistant using LangGraph & HITL

1. Module Structure

```
src/
 ├── ingestion/
 ├── retrieval/
 ├── llm/
 └── workflow/
```

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2. Ingestion Layer

loader.py

- Loads healthcare PDFs

chunker.py

- Splits text into chunks
- Chunk Size: 500
- Chunk Overlap: 100

pipeline.py

- Executes ingestion workflow

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3. Retrieval Layer

vector_store.py

- Creates embeddings
- Stores vectors in ChromaDB

retriever.py

- Retrieves top relevant chunks

Embedding Model:

all-MiniLM-L6-v2

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4. LLM Layer

llm.py

- Uses Groq LLM
- Generates grounded responses

- Handles refinement prompts

Model Used:

llama-3.1-8b-instant

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5. Workflow Layer

state.py

Stores:

- query
- answer
- feedback
- chat_history
- retry_count

nodes.py

Main Node:

chat_node()

Responsibilities:

- Retrieve chunks
- Generate answers
- Handle feedback refinement
- Update conversational memory

graph.py

Creates LangGraph workflow:

START → chat_node → END

Uses:

MemorySaver()

for state persistence.

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6. HITL Workflow

Generate Answer
↓
Human Review
↓

If NO → Refine Answer
↓
Retry Limit Check

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7. Memory Handling

- Sliding window conversational memory
- Stores last 3 conversations

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8. Testing

Tested using:

- Healthcare queries
- Follow-up questions
- HITL refinement
- Unknown queries

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9. Future Improvements

- Better conversational retrieval
- Query rewriting
- Advanced memory
- Production deployment